

Jul 28, 2021

CELL STORAGE-02-Freezing and Thawing Protocol for Suspension Cell Lines

DOI

<https://dx.doi.org/10.17504/protocols.io.bvwxn7fn>

Marco Cosentino¹, Massimiliano Legnaro¹, Emanuela Rasini¹, Mariagiulia Albizzati¹, Marco Ferrari¹,
Franca Marino¹, Alessandra Luini¹

¹Center for Research in Medical Pharmacology, University of Insubria (Varese, Italy)

Center for Research in Medical Pharmacology
Tech. support email: farmacologia.medica@uninsubria.it



Farmacologia Medica

OPEN  ACCESS



DOI: <https://dx.doi.org/10.17504/protocols.io.bvwxn7fn>

Protocol Citation: Marco Cosentino, Massimiliano Legnaro, Emanuela Rasini, Mariagiulia Albizzati, Marco Ferrari, Franca Marino, Alessandra Luini 2021. CELL STORAGE-02-Freezing and Thawing Protocol for Suspension Cell Lines. **protocols.io**
<https://dx.doi.org/10.17504/protocols.io.bvwxn7fn>

License: This is an open access protocol distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working

We use this protocol and it's working

Created: June 17, 2021

Last Modified: May 18, 2026

Protocol Integer ID: 50871

Keywords: protocol for suspension cell line, suspension cell line, cell morphology evaluation, cells quality, freezing procedure, cells viability, cells morphology, cell line, thawing procedure, cells damage, quality test cell, cell, freezing process, shaped cells morphology, increasing cells viability, useful to minimizes ice crystals formation, freezing, dmso toxic effect, thawing protocol, following cell line, t25 cm2 culture flask, freezing time, minimizes ice crystals formation, using optical microscopy technique, optical microscopy technique, expected freezing time

Abstract

In our institute, cell lines are stored after freezing procedure that provides for a constant temperature lowering by 1 degree/minute. This procedure is useful to minimize ice crystals formation during freezing process, this allows to reduce the cells damage and, in turn, increasing cells viability after thawing. After storage at -80°C or in liquid nitrogen (in relation to subsequent applications or the expected freezing time) cells are thawed rapidly in a water bath at 37°C and the cryopreserving-medium is immediately removed by gentle centrifugation in order to reduce as soon as possible the DMSO toxic effect.

Currently, in our institute, the freezing/thawing procedure is validated for the following cell lines:

- **PC-12**, Rat adrenal pheochromocytoma, small irregularly shaped cells morphology, suspension growth mode.

After freezing/thawing procedure, cells quality must be assessed by cell morphology evaluation (using Optical Microscopy technique) and time required to reach confluence in the T25 cm² culture flask, which must not exceed the times indicated in the **Table 1**.

After quality test cells must be resuspended according to specific protocols (for details see Table 2).

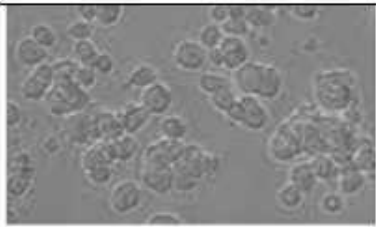
Cell lines	Time to Reach Confluence	Reference for morphology evaluation
PC12	72 hours	

Table 1

Cell-lines	Culture-medium	Applications-post-thawing	Our-publication-using-this-applications
PC12	RPMI-1640+5%FBS+10%FBS+100 U/ml penicillin/streptomycin	Cell-cultures	Cosentino-et-al.,2019

Table 2

List of published work by our institute that using this procedure

- Cosentino M, Marino F, Rasini E, Legnaro M, Bombelli R, Luini A, Pacchetti B. **Improved solubility and increased biological activity of NeoSol™ RCL40, a novel Red Clover Isoflavone Aglycones extract preparation.** Biomed Pharmacother. 2019 Mar;111:91-98. doi: 10.1016/j.biopha.2018.12.065. Epub 2018 Dec 19. PMID: 30579257.



Materials

Reagents

- **FBS** catalog number: ECS0180L-500 ml, Euroclone
- **RPMI 1640** catalog number: ECM0495L-500 ml, Euroclone
- **FHS** catalog number: ECS0090L-500 ml, Euroclone
- **Penicillin/streptomycin** catalog number: ECB3001D-500 ml, Euroclone
- **DMSO** catalog number 276855, Sigma-Aldrich

Sterile plastic disposable:

- **Cryogenic vials** catalog number: CC430659, Corning
- **T25 cm² culture flasks for suspension cells** catalog number: CC431463, Corning

Instrumentation required:

- Laminar flow hood
- Centrifuge
- -80°C freezer



CELL FREEZING PROCEDURE FOR SUSPENSION CELLS (PC-12)

- 1 At confluence remove cell suspension from the T25 cm² culture flask and recover in 15 mL conical tube.
- 2 Centrifuge for 00:05:00 at 200 x g Room temperature .
- 3 Remove supernatants and resuspend the cell pellet in 1 mL of cryopreserving medium (90% FBS + 10% DMSO).
- 4 Immediately aliquot 1 mL of cell suspension into a cryogenic vial.
- 5 Place the cryogenic vial(s) in cotton wool, and then place it in a polystyrene box.
- 6 Put the polystyrene box in -80°C freezer.

Equipment

new equipment

NAME

-80°C Freezer

TYPE

Eppendorf

BRAND

B U9230-0001

SKU

- 7 After a minimum of 24:00:00 transfer the cryogenic vials into a box for storage in -80°C freezer.

1d



CELL THAWING PROCEDURE

5m

- 8 Remove cryogenic vial from -80°C and keep it at  37 °C in water bath, this procedure should take approximately 1-2 minutes or until ampoule is slightly thawed.
- 9 Clean the cryogenic vial outside with 70% EtOH, open the cryogenic vial and add  1 mL of prewarmed complete medium for specific cells lines (**Table 2**).
- 10 Transfer the cell suspension into a 15 mL conical tube containing  10 mL of prewarmed complete medium for specific cells lines.
- 11 Centrifuge for  00:05:00 at  200 x g  Room temperature .
- 12 Resuspend the cells into appropriate cell culture medium (**Table 2**) and start quality control procedure (**Table 1**).

5m